

PHY101 QUIZ 4

ORANGE MONKEY TEAM

You are being informed that the 4th online quiz of Physics (PHY101) is going to held on 07th March, 2022 to 08th March, 2022. The quiz will consist of Multiple Choice Questions (MCQs) and cover Lessons 23 to 43.

NOTE

**“QUIZ ATTEMPT KRNY SEY PHELY
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LYNA BCUZ MCQS NEW ARAHY HEIN
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SEARHC KRLYNA JOO YAHN SEY NA
MILEIN >>>JAZAKALLAH”**

1. The reason there are two slits, rather than one, in a young's experiment is:

To create a path length difference....confirm from net

2. According to the kinetic theory of gasses, the pressure of a gas is due to:

Change of momentum of molecules as the strike the well....confirm from net

3. What is the name of the relationship between the peak wavelength for a blackbody radiator and its temperature.

Wien's law

4. What is the unit of magnification factor?

No units

5. The special theory of relativity is applicable to the object moving with maximum velocity equal to

More than speed of light

6. A free electron and a free proton have the same kinetic energy . this means that, compared to the matter wave associated with the proton, the matter wave associated with the electron has:

A longer wavelength and the same frequency

7. No entropy change is associated with:

Adiabatic process

8. Electromagnetic waves emitted from antenna are:

Transverse

9. A virtual image is one

From which light rays diverge but do not pass through

10. A detector is used to detect the speed at a particular wave if the frequency of an electromagnetic signal is 400 MHz, the speed define the detector is

300000 km/s

11. Evidence that molecules of a gas are in constant motion is:

Two gases interdi? Use quickly

12. Of the following which is the best evidence for the wave nature of matter?

The photoelectric e? ect

13. For holography we use a beam of:

Laser

14. Interference of light is evidence that:

Light is a wave phenomenon

15. Which of the following types of electromagnetic radiation travels at the greatest speed in vaccum?

(Radio, gamma, x)rays .. all options....confirm

16. The plants look green. It is because they have chlorophyll in them. Chlorophyll makes leaves green to absorb_____.

Green only

17. The most refined form of matter is_____.

Light

18. The known laws of electromagnetism before James Clerk Maxwell were . Except

Faraday's law

19. Which of the following electromagnetic radiations has photons with the greatest energy?

Radio waves

20. The primary of a 3:1 step-up transformer is connected to a source and the secondary is connected to a resistor R. The power dissipated by R in this situation is P. If R is connected directly to the source it will dissipate a power of:

$P/9$

21. The quantum number m_s is most closely associated with what property of the electron in an atom?

Energy

22. The special theory of relativity is based on:

Two postulates

23. The center of Newton's ring is dark due to:

Destructive interference

24. Which electromagnetic radiation transmits the highest photon energy/

Gamma rays

25. Which methods of heat transfer can take place in vacuum?

Radiation

26. An electron is moving at $0.6c$. If we calculate its kinetic energy using $(1/2)mv^2$. We get a result that is:

About 28% too low

27. The wavelength of light beam A is twice the wavelength beam B. the energy of a photon in beam A is:

Half the energy of a photon in beam b

28. The lanz's law refers to :

Induced current

29.

30. Electric energy is measured :

Watt....confirm from ent

31. The lenz's law refers to :

Induced current

32. The films formed due to detergents(soap/shampoo) are due to

interference

33. A free electron and a free proton have the same kinetic energy . this means that, compared to the matter wave associated with the proton, the matter wave associated with the electron has:

A longer wavelength and a smaller frequencyconfirm from net

34. Of the following. Compton scattering from electrons is most easily observed for:

X rays

35. Of the following which might NOT vanish over one cycle of a cyclic process?

The work done by the substance....confirm from net

36. The most refined form of matter is:

Light

37. How could the unit of potential difference, the volt, also be written?

J/C

38. In Compton scattering from stationary particles the maximum change in wavelength can be made smaller by using.

More massive particles

39. Hotness or coldness of an object is represented in terms of:

Temperature

40. The unit of thermal conductivity might be:

Cal/(cm*s*C?)

41. During the time that latent heat is involved in a change of state.

The temperature does not change

42. A laser in a compact disc player generates light that has a wavelength of 780 nm in air. The light then enters into the plastic of a CD. If the index of refraction of plastic is 1.55, the speed of this light once enter the plastic is _____.

1.94×10^8 m/s

43. Antiparticle of electron is

Positron.....confirm from net

44. When some dielectric is inserted between the plates of a capacitor then capacitance

Increases....confirm from net

45. The centre of Newton ring is dark due to:

Destructive interference

46. If two objects are in thermal equilibrium with each other:

They cannot be at different temperatures....confirm from net

47. The graphical representation of Ohm's law is

Straight line.....confirm from net

48. A particular device transmitted a signal in the form of electromagnetic waves of frequency 7.5×10^8 Hz. What is the wavelength of the signal?

0.4m

41. The resolving power of a telescope can be increased by

Increasing the lens diameter

49. Which physical quantity is produced by a calculation where a charge is multiplied by a potential difference (p.d)?

Energy....confirm form net

50. A transparent refracting medium bounded by the two curved surfaces called:

Prism....confirm from net

51. It is more difficult to measure the coefficient of volume expansion of a liquid than that of a solid because?

The containing vessel also expands....confirm form net

52. The product of resistance and capacitance is called

Time....confirm from net

53. The potential difference between the ends of a conductor is 12 V. How much electrical energy is converted to other forms of energy in the conductor when 100C of charge flows through it?

1200....confirm form net

54. When we accelerate the charge, which types of waves are produced?

Electromagnetic waves

55. Due to relative motion of observer and the frames of reference of events, time always:

slows itself

56. The allowed energy states of atoms or molecules can be described by a set of _____.

Quantum numbers.....confirm

57. Probability values are assigned on a scale of

0 to 1....confirm

58. The magnitude of the momentum of a particle can never exceed:

E/c , where E is its total energy

59.

60. The ratio of the size of image to the size of object.

Angular magnification

61. When white light passes through a prism, it splits into _____ colors.

7.....confirm from net

62. A thermometer indicates 98.6°C. it may be:

IN a cup of hot tea

63. Tesla is a unit of

Magnetic flux

64. A light entering into glass prism from air does not give change in its:

Frequency

65. The total energy in an LC circuit is 5.0×10^{-6} J. If $L = 25$ mH the maximum current is:

20mA

66. Electromagnetic waves do not transport:

Charge

67. A heat of transformation of a substance is:

The energy per unit mass absorbed as heat during a phase transformation

68. Which of the following is true of all virtual images?

None of these

69. $2d \sin \theta = m\lambda$ is called

bragg equation

70. No lens is perfect because _____.

They suffer from aberration

71. A basic postulate of Einstein's theory of relativity is

The laws of physics must be the same of observers moving with uniform velocity relative to each other.

72. Evidence that a gas consists mostly of empty space is the fact that

Heating a gas increases the molecular motion

73. The number of turns becomes double, but length remain same, then magnetic field in the solenoid become.

Double

74. Which of the following electromagnetic radiations has photons with the greatest energy?

X rays

75. When two objects come to common temperature, the body is said to be in:

Thermal equilibrium....confirm from net

76. If electric and gravitational force on an electron in a uniform electric field balance each other, then the intensity of electric field will be:

Mg/q

77. Electromagnetic waves do not transport:

Charge

78. During a slow adiabatic expansion of a gas

No energy enters or leaves as heat

79.

80. A free electron and a free proton have the same kinetic energy. This means that, compared to the matter wave associated with the proton, the matter wave associated with the electron has.

A longer wavelength and the same frequency

81. The quantity $\Delta(V)/\Delta(r)$ is called

Potential gradient

82. If two objects are in thermal equilibrium with each other.

They cannot be at different temperatures

83. A consequence of Einstein's theory of relatively is:

Moving clocks run more slowly than when they are at rest

84. In order that a signal process is both isothermal and isobaric

A change of phase is essential

85. The specific heat of a substance is:

The amount of heat energy per unit mass to raise the temperature of the substance by 1°C

86. The electron states that constitute a single shell for an atom all have:

The same value of n

87. An isothermal process for an ideal gas is represented on a p-V diagram by:

A portion of a hyperbola

88. The concept of an electric field lines is introduced by:

Faraday

89. The current in a car headlamp is 2.0A. The headlamp is switched on for 30 minutes. How much charge passes through the headlamp?

360 C

90. In some movies, you sometimes see an actor looking in a mirror and you can see his face in the mirror. During the filming of this scene, what does the actor see in the mirror?

The movie camera

91. The Pauli exclusion principle is obeyed by:

All particles with spin quantum numbers of $\frac{1}{2}$

92. The focal length of a convex lens is:

Positive

93. An electron ($m = 9.11 \times 10^{-31} \text{ kg}$) has a speed of $0.95c$. Its kinetic energy is;

$1.8 \times 10^{-13} \text{ J}$

94. An electron ($m = 9.11 \times 10^{-31} \text{ kg}$) has a speed of $4.0 \times 10^8 \text{ m/s}$. Its kinetic energy is:

$1.8 \times 10^{-13} \text{ J}$

95. An electron ($m = 9.11 \times 10^{-31} \text{ kg}$) has a momentum of $4.0 \times 10^{-22} \text{ kg}\cdot\text{m/s}$. Its kinetic energy:

$6.3 \times 10^{-14} \text{ J}$

96. Force on a charged particle is zero when projected at angle with the magnetic field

0 degree

97. Heat is:

Energy transferred by virtue of a temperature difference

98. When a charge is projected perpendicular to a uniform magnetic field, its path is:

Circular

99. If potential difference across two plates of a capacitor is doubled, then energy stored in it will be:

Four times

100. charge carriers are.

In case of metallic conductors, the

Electrons

101. degrees on the:

Room temperature is about 20 to 22

Celcius scale

102. degrees on the

Room temperature is about 68 to 72

Fahreheit scale

103. an observer on Earth, the distance from earth star measured by an observer in a moving spaceship would seem:

As compared to distance measured by

Same

104. empty space is the fact that

Evidence that a gas consists mostly of

The density of a gas becomes much greater when it is
liquefied....confirm

105.

106.

For an electron, the rest mass energy is:

0.511 MeV.....confirm from net

107.

Antiparticle of electron is:

Position

108.

According to the first law of thermodynamics, applied to a gas, the increase in the internal energy during any process

net

Equals the heat input plus the work done on the gas....confirm from

109.

The approach to personality stresses that people possess a basic goodness, and have a natural tendency to grow to higher levels of functioning is.

humanistic....confirm from net

110.

In a photoelectric experiment at a frequency above cut off, the stopping potential is proportional to :

The energy of the most energetic electron after it is ejected....confirm from net

111.

There is a temperature at which the reading on the kelvin scale is numerically:

Equal to that on the Celsius scale

112.

When you stand in front of a plane mirror, your image is:

Virtual, erect and the same size as you

113.

When a ray of light travels from a lesser denser medium to denser medium then _____.

It bends towards the normal

114.

Entropy measure the:

Energy of a system

115.

The theoretical upper limit for the frequency of electromagnetic waves is:

There is no upper limit

116. The special theory of relativity is applicable to the object moving with maximum velocity equal to:

Speed of light

117. According to the special theory of relativity:

All forms of energy have mass-like properties

118. In constructing a thermometer it is necessary to use a substance that:

Undergoes some change when heated or cooled

119. A particle with rest mass m moves with speed $0.6c$. its kinetic energy is:

$0.25 mc^2$

120. J.J Thompson's measurement of e/m for electrons provides evidence of the:

Particle nature of matter

121. Polarization means _____.

To alter the wavelength of the light

122. Emission of electron by metals on heating is called

Thermionic emission

123. The specific heat of a substance is: the amount of heat energy per unit mass to raise the temperature of the substance by 1°C

124. An erect object is in front of a convex mirror a distance greater than the focal length. The image is:

Virtual, erect, and smaller than the object.....confirm from net

125. The appearance of colors in thin film is due to:

Interface.....confirm from net

126. A student rubs a rod held in his hand.
Which action causes the rod to gain a large electrostatic charge?

Rubbing a polythene rod with a woolen duster

127. Two parallel straight wires carrying
current in same direction

Attract each other

128.

129.

One tesla is equal to

N/Am.....confirm from net

130.

constant is:

The process for which entropy remains

A reversible process

131.

due to:

The appearance of colors in thin film is

interference

132.

radiation

Greenhouse gases absorb _____

Infrared

133.

frequency of electromagnetic waves is:

The theoretical upper limit for the

There is no upper limit

134.

Heat has the same units as:

Work

135.

given by:

Capacitance of a capacitor in vacuum is

$\epsilon_0 A / d$

136.

the principal quantum number:

An electron in a K shell of an atom has

$N = 1$

137. When an electromagnetic wave meets a reflecting surface, the direction taken by the reflected wave is determined by:

The angle of incidence

138. Capacitive reactance X_c is

$\frac{1}{2(\pi)fc}$

139. Bending of light around the edges of an obstacle called:

Diffraction

140. In RL series circuit phase angle is given by:

$\tan^{-1}(\omega L/R)$

141. Interface of light is evidence that:

Light is a wave phenomenon

142. In a photoelectric effect experiment at a frequency above cut off the number of electrons ejected is proportional to:

The number of photons that hit the sample.....confirm from net

143. In a photoelectric effect experiment the stopping potential is:

The electric potential that causes the electron current to vanish...confirm from net

144. In an adiabatic process:

The work done by the environment on the system equals the change in internal energy....confirm from net

145. A free electron and a free proton have the same kinetic energy. This means that, compared to the matter wave associated with the proton, the matter wave associated with the electron has:

A longer wavelength and the same frequency...confirm from net

146. Polarization experiments provide evidence that light is:

A transverse wave....confirm from net

147. If light of low wavelength is used in young's double slit experiment, then width of the fringe will:

Decreases....confirm from net

148. Lenz's law is in accordance with law of conservation of

Energy....confirm from net

149. AN LC series circuit with an inductance L and a capacitance C has an oscillation frequency f. Two inductors, each with inductance L, and two capacitors, each with capacitance C, are all wired in series and the circuit is completed. The oscillation frequency is:

F

150. The known laws of electromagnetism before James Clerk Maxwell were following EXCEPT:

Faraday's Law

151. A particle having $2e$ charge falls through a potential difference of 5V energy acquired by it is.

10eV

152. According to relatively theory a particle of mass m with a momentum of $2mc$ has a speed of:

$c/2$

153. If the kinetic energy of a free particle is much less than its rest energy then its kinetic energy is proportional to:

The square of the magnitude of its momentum

154. The two metallic strips that constitute some thermostats must differ in:

Coefficient of linear expansion

155. Phase difference between V and I of an AC through resistor is:

0 degree

156. Average value of current and voltage over complete cycle is:

zero

157. A detector is used to detect the speed of a particular wave. If the frequency of an electromagnetic signal is 400 MHz, the speed detected by detector is _____.

300000 km/s

158. Inductive reactance X_L of an inductor is:

$2\pi f L$confirm from net

159. An inductor may store energy in its:

Magnetic field....confirm from net

160. If velocity of a conductor moving through a magnetic field B is made zero then motional emf is:

$-vBL$

